

The Problem with RAG in Healthcare: Ambiguity In, Ambiguity Out

RAG can make AI much better at finding information. But what happens when the information it finds is inconsistent?

If ten different RAG → AI applications encounter the same conflicting healthcare information from multiple providers' healthcare information systems, what are the chances they will resolve those conflicts the same way and deliver the same conclusions to the physician?

Much of today's thinking can be simplified as:

Multiple Healthcare Information Systems → RAG → AI → Human Decision

Each application may have to retrieve, interpret, and reconcile the same conflicting information before addressing the problem at hand. That creates risks of inconsistent results while adding processing cost and response time.

The same information may be retrieved, processed, and reconciled again for the next application—creating another opportunity for a different conclusion.

It also makes each application harder to build and validate. One process must reconcile conflicting evidence, decide what to trust, identify what matters now, and organize it for the physician.

What if we modularized the architecture and freed each application from resolving the patient's history?

Healthcare Data → Extract & Structure → Reconcile & Integrate → Longitudinal Knowledge

Extraction and reconciliation become persistent, independently testable functions. Resolve what can be resolved, preserve what cannot, and allow providers and patients to correct questionable information.

Then every downstream application starts from the same longitudinal knowledge:

Longitudinal Knowledge → Retrieve → AI → Human Decision

The delivery AI can focus on the problem at hand rather than reconstructing and reconciling the patient's history each time.

Resolve once. Correct when necessary. Use many times.

This distinction became a central idea in *The Missing Healthcare Opportunity*, which explores the architecture, conceptual data model, AI roles, and potential benefits of creating a trustworthy longitudinal patient history.

The book concludes with a proposal for a six-person, 60-day feasibility assessment to determine whether the potential value is significant, the architecture is technically feasible, and the major barriers are reasonably addressable.

The Right Information. In Time. Every Time.